

Could AI use ocean water from rising sea levels to cool their machines instead of tap water?

Posted by u/Switchell22 • 0 points • 11 comments

The main concern I hear about AI even more than the ethical implications and debate on whether or not AI is fair use or not is the environmental impact. I believe I've heard a single response from ChatGPT consumes an entire bottle of water.

Sea levels are rising and are a very real threat to coastal cities. If AI consumes that much water, couldn't we just pump in water directly from the oceans and consume that water for coolant instead?

I don't imagine this would make a massive impact on rising sea levels, but a small impact is still an impact. And it would hopefully drive the costs of tap water back down.

Comments

u/III-Salamander • 1 points • 2024-10-31 18:27 UTC

>I believe I've heard a single response from ChatGPT consumes an entire bottle of water.

Whoever told you that is speaking nonsense. Computers use electricity, not water.

u/Switchell22 • 0 points • 2024-10-31 19:21 UTC

Well yes obviously electronics run on electricity. Nobody is saying AI servers somehow magically use water instead. The concerns are how much water is evaporated in the cooling process to prevent them from overheating.

But that part I don't know how to fact check myself. Water cooling is a thing yeah, I use it, but I've never heard of needing so much water to cool a machine that it actively evaporates it.

But my question is working off the assumption that those metrics people frequently refer to are accurate.

u/III-Salamander • 1 points • 2024-10-31 20:25 UTC

Water evaporating won't solve rising ocean levels because evaporated water doesn't disappear.

u/Ridley_Himself • 2 points • 2024-10-31 18:26 UTC

Water is not consumed. Water cooling either used a fixed amount of water in a closed system or releases water back into the environment, likely as steam. Any water released into the environment ultimately returns to the ocean.

Seawater is a poor choice of coolant because salt is quite corrosive and the process of removing that salt is expensive.

u/bangbangracer • 1 points • 2024-10-31 18:07 UTC

Microsoft actually was experimenting with underwater data centers. They filled a bunch of submersible pods with servers, then sank them off the west coast. It actually was a viable concept, but decided against because of salt concerns and difficulty performing on-site maintenance if needed.

What you are describing isn't really apples to apples though and isn't an option.

u/Dave_A480 • 1 points • 2024-10-31 18:07 UTC

You can use whatever you want to engineer for.....

Eg, ships use ocean water for coolant but the system has to be designed to handle the corrosive effects of salt water....

With that said, it's not like cooling water from a data center can't be treated and reused like any other wastewater....

It doesn't just vanish.

The larger issue is that the states best suited water wise for this sort of thing are the Midwestern ones (pump it out of the Great Lakes, use it to cool data centers, treat it and put it back in the lake) with very business-hostile legal environments, as a legacy of their part on the old industrial economy....

u/JustSomeGuy_56 • 1 points • 2024-10-31 18:37 UTC

>With that said, it's not like cooling water from a data center can't be treated and reused like any other wastewater....

Circa 1979 I worked for big insurance company that used the water which cooled the mainframes to heat the building, We got some kind of award from the Dept of Energy.

u/Orangeshowergal • -3 points • 2024-10-31 17:57 UTC

The "liquid" in liquid cooling isn't water lol

u/1king-of-diamonds1 • 1 points • 2024-10-31 17:55 UTC

There's been a few decent AMAs on this, the general gist of that salt water contains too many minerals for prolonged use in metal. I did remember hearing about physical underwater data centers but I don't imagine those were very practical either

u/StripedMaple-1238 • 2 points • 2024-10-31 17:54 UTC

Salt water is corrosive and much more difficult to work with in the context of cooling.

It's not that it's impossible, just that it's much more expensive than using tap water.

By the way, this is generally the case when we're talking about water shortages. It's not a shortage of actual water, because the ocean is massive. It's a shortage of easily and (relatively) cheaply accessible fresh water. Areas like Arizona would have to invest massive amounts of money into infrastructure like water pipelines or trucks to meet their current water needs without groundwater or the Colorado River.

u/Inevitable-Regret411 • 6 points • 2024-10-31 17:51 UTC

Salt water makes poor coolant because it's corrosive. Saltwater rusts metal at a fantastic rate, the pipes would need replacing much faster and would clog more frequently. Plus the amount of water consumed by AI isn't anywhere comparable to the threat posed by rising sea levels. It would be like trying to bail out the Titanic with a shot glass. Better than nothing yes, but it's not going to help anything, and instead of bailing out you could be doing something useful like handing out lifejackets.

u/ScheduleOk3809 • 12 points • 2024-10-31 17:46 UTC

Water does not get consumed by cooling. This would do nothing to stop rising sea levels.

u/[deleted] • 1 points • 2024-10-31 17:51 UTC

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u/Corgipantaloonss • 2 points • 2024-10-31 17:45 UTC

At this point the cost of filtering and removing the salt and other crap outweighs the benefit. That takes a large amount of resources to properly do, especially if it's being used for anything mechanical.

Otherwise we wouldn't have the issue of running out of potable water in the first place.